



Ensemble Machine Learning Approaches for Predicting Academic Distress and Mental Health Risk in Higher Education

A Multi-Dataset Comparative Study

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The Problem

Why Does This Matter?



Rising Crisis

Stress, anxiety & depression are among the top reasons students drop out globally. [1, 2]



Missed Early Signs

Most at-risk students never seek help systems fail to flag them in time. [1]



Unused Data

Institutions collect rich student data but rarely use it for early intervention. [3, 4]

This Study's Answer:

Apply machine learning to predict academic distress & mental health risk before it escalates.

Research Gap

Where the Literature Falls Short



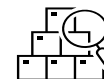
Existing Studies Focus

- Single mental health outcomes {5}
- Heavy reliance on complex ensemble/deep learning models [5, 6]



Gap Identified

- Limited comparative analysis across stress & depression datasets
- Lack of emphasis on interpretability for educational deployment [3]



This Study

- Comparative multi-dataset approach
- Robustness-focused evaluation
- Emphasis on interpretable ML models

The Data

Dataset A — Depression

Binary Classification (Depressed: Yes / No)

- Depression (target)
- Academic Pressure
- Financial Stress
- Suicidal Thoughts
- Sleep Duration
- Dietary Habits
- CGPA
- Work Hours
- Gender

Dataset B — Stress

Multiclass Classification (Low / Medium / High)

- Stress Level (target)
- Anxiety Level
- Sleep Quality
- Bullying
- Peer Pressure
- Social Support
- Academic Performance
- Future Career Concerns

Variables include psychological, physiological, social factors. Both datasets sourced from Kaggle publicly available and reproducible. [9, 10]

Methodology



Data Preprocessing

Remove noise · encode categories · discretise continuous vars · mode imputation for missing values [8]



Three ML Models

Logistic Regression · Decision Tree · Random Forest — selected to compare linear vs. ensemble approaches [6, 7]



Evaluation

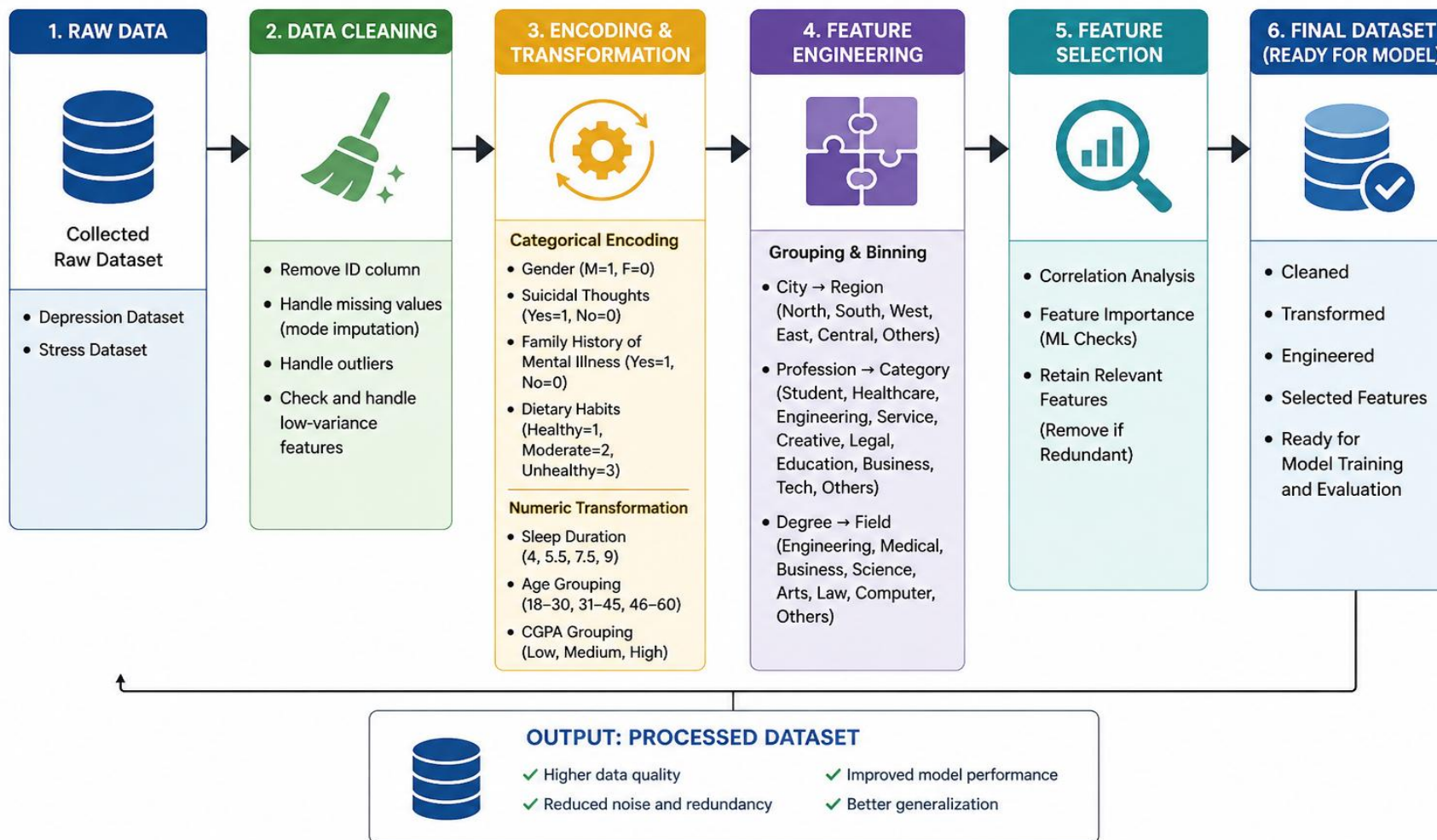
Accuracy · Precision · Recall · F1-Score across both datasets independently [8]



Robustness Testing

Missing data · Outlier sensitivity · Multicollinearity — confirming real-world viability [3]

DATA PREPROCESSING PIPELINE



Before Preprocessing



AB25																			
	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
1	id	Gender	Age	City	Profession	Academic	Work Pres	CGPA	Study Sati	Job Satisf	Sleep Duration	Dietary Ha	Degree	Have you e	Work/Stuc	Financial	Family His	Depression	
2	2	Male	33	Visakhapatna	Student	5	0	8.97	2	0	'5-6 hours'	Healthy	B.Pharm	Yes	3	1	No	1	
3	8	Female	24	Bangalore	Student	2	0	5.9	5	0	'5-6 hours'	Moderate	BSc	No	3	2	Yes	0	
4	26	Male	31	Srinagar	Student	3	0	7.03	5	0	'Less than 5 hou	Healthy	BA	No	9	1	Yes	0	
5	30	Female	28	Varanasi	Student	3	0	5.59	2	0	'7-8 hours'	Moderate	BCA	Yes	4	5	Yes	1	
6	32	Female	25	Jaipur	Student	4	0	8.13	3	0	'5-6 hours'	Moderate	M.Tech	Yes	1	1	No	0	
7	33	Male	29	Pune	Student	2	0	5.7	3	0	'Less than 5 hou	Healthy	PhD	No	4	1	No	0	
8	52	Male	30	Thane	Student	3	0	9.54	4	0	'7-8 hours'	Healthy	BSc	No	1	2	No	0	
9	56	Female	30	Chennai	Student	2	0	8.04	4	0	'Less than 5 hou	Unhealthy	'Class 12'	No	0	1	Yes	0	
10	59	Male	28	Nagpur	Student	3	0	9.79	1	0	'7-8 hours'	Moderate	B.Ed	Yes	12	3	No	1	
11	62	Male	31	Nashik	Student	2	0	8.38	3	0	'Less than 5 hou	Moderate	LLB	Yes	2	5	No	1	
12	83	Male	24	Nagpur	Student	3	0	6.1	3	0	'5-6 hours'	Moderate	'Class 12'	Yes	11	1	Yes	1	
13	91	Male	33	Vadodara	Student	3	0	7.03	4	0	'Less than 5 hou	Healthy	BE	Yes	10	2	Yes	0	
14	94	Male	27	Kalyan	Student	5	0	7.04	1	0	'Less than 5 hou	Moderate	M.Tech	No	10	1	Yes	1	
15	100	Female	19	Rajkot	Student	2	0	8.52	4	0	'Less than 5 hou	Unhealthy	'Class 12'	No	6	2	Yes	0	
16	103	Female	19	Kalyan	Student	5	0	5.64	5	0	'Less than 5 hou	Moderate	'Class 12'	Yes	4	5	Yes	1	
17	106	Male	29	Srinagar	Student	3	0	8.58	3	0	'More than 8 hou	Moderate	M.Tech	Yes	10	2	Yes	1	
18	120	Male	25	Nashik	Student	5	0	6.51	2	0	'Less than 5 hou	Unhealthy	M.Ed	Yes	2	5	Yes	1	
19	132	Female	20	Ahmedabad	Student	5	0	7.25	3	0	'5-6 hours'	Healthy	'Class 12'	Yes	10	3	No	1	

After Preprocessing

AutoSave ☐ Off    sdd_preprocessed.csv

File Home Insert Draw Page Layout Formulas Data Review View Automate Help

Clipboard: Paste, Cut, Copy, Format Painter

Font: Aptos Narrow, 11, Bold, Italic, Underline, Text Color, Background Color

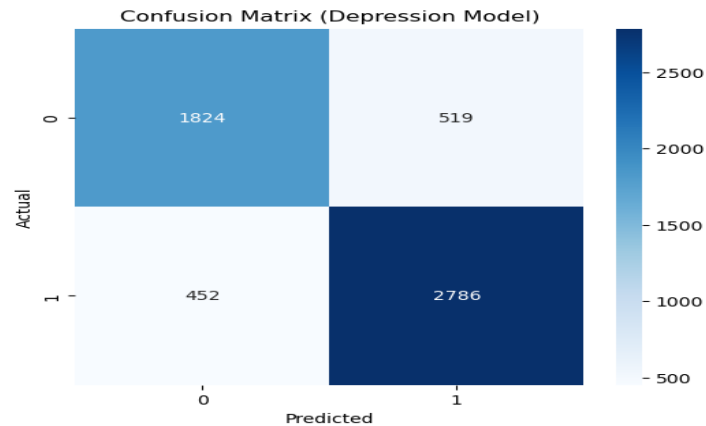
Alignment: Left, Center, Right, Indent, Wrap Text, Merge & Center

Number: General, Percentage, Currency, Date, Time, Text, Scientific

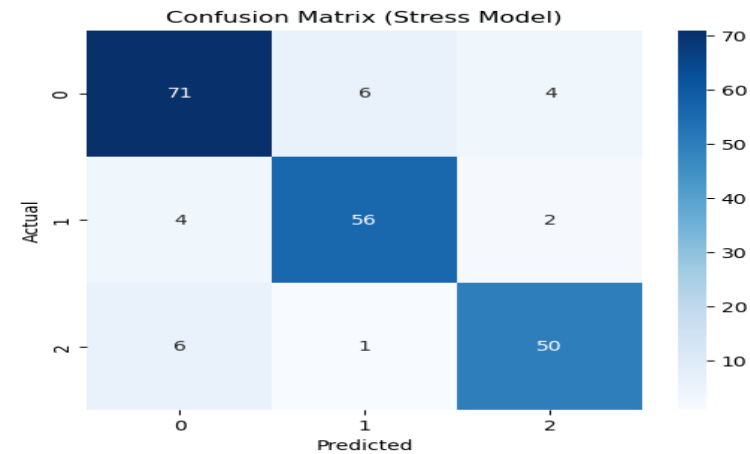
Styles: Normal, Good, Bad, Neutral

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
1	Gender	Age	City	Profession	Academic	Work Pres	CGPA	Study Sati	Job Satisf	Sleep Dur	Dietary Ha	Degree_Fi	Suicidal T	Work/Stuc	Financial	FM Mental	Depression	
2	0	2	2	1	5	0	3	2	0	2	1	2	1	1	1	0	1	
3	1	1	2	1	2	0	1	5	0	2	2	4	0	1	2	1	0	
4	0	2	1	1	3	0	2	5	0	1	1	5	0	3	1	1	0	
5	1	1	4	1	3	0	1	2	0	3	2	7	1	1	5	1	1	
6	1	1	6	1	4	0	3	3	0	2	2	1	1	1	1	0	0	
7	0	1	3	1	2	0	1	3	0	1	1	8	0	1	1	0	0	
8	0	1	3	1	3	0	3	4	0	3	1	4	0	1	2	0	0	
9	1	1	2	1	2	0	3	4	0	1	3	8	0	1	1	1	0	
10	0	1	3	1	3	0	3	1	0	3	2	8	1	3	3	0	1	
11	0	2	3	1	2	0	3	3	0	1	2	6	1	1	5	0	1	
12	0	1	3	1	3	0	2	3	0	2	2	8	1	3	1	1	1	
13	0	2	3	1	3	0	2	4	0	1	1	1	1	3	2	1	0	

Validation



- 84.7% of depressed students correctly identified
- False negatives kept low — critical in mental health contexts



- Errors only between neighbouring stress levels, no major misclassification
- All three stress classes reliably detected

Both models performance holds under missing data and outliers

Key Results

Stress

```
results_df = pd.DataFrame({
    "Logistic Regression": lr_results,
    "Decision Tree": dt_results,
    "Random Forest": rf_results
}).T

print(results_df)
```

	Accuracy	Precision	Recall	F1 Score
Logistic Regression	0.915	0.915643	0.915	0.914935
Decision Tree	0.890	0.891106	0.890	0.890156
Random Forest	0.885	0.885020	0.885	0.884972

Depression

```
results_df = pd.DataFrame({
    "Logistic Regression": lr_results,
    "Decision Tree": dt_results,
    "Random Forest": rf_results
}).T

print(results_df)
```

	Accuracy	Precision	Recall	F1 Score
Logistic Regression	0.833005	0.832451	0.833005	0.832454
Decision Tree	0.757212	0.757404	0.757212	0.757304
Random Forest	0.826017	0.825518	0.826017	0.825646



Best Performer

Logistic Regression tops both datasets — 91.5% stress, 87.3% depression.



Why Simpler Wins

This demonstrates that effective preprocessing and feature engineering can reduce the need for model complexity [6, 6]



Robust

Performance holds under missing data, outliers & multicollinearity. [7]

$r = 0.7569$

Career concerns $\leftrightarrow$ stress (top correlation)

$r = -0.7646$

Sleep quality $\leftrightarrow$ stress (protective factor)

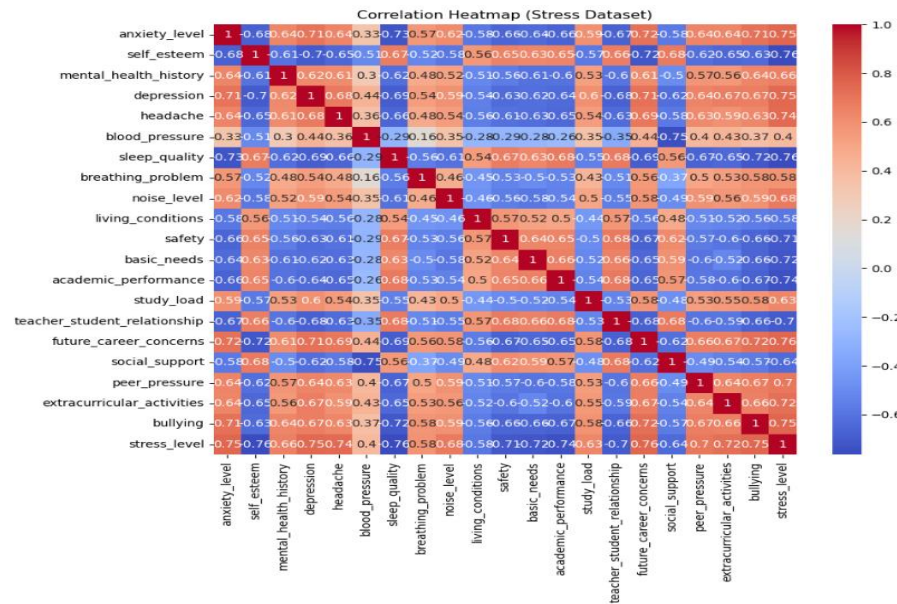
$r = 0.5463$

Suicidal thoughts $\leftrightarrow$ depression (top link)

What Drives Stress vs Depression



Stress Predictors - Correlation



Future career concerns

0.757

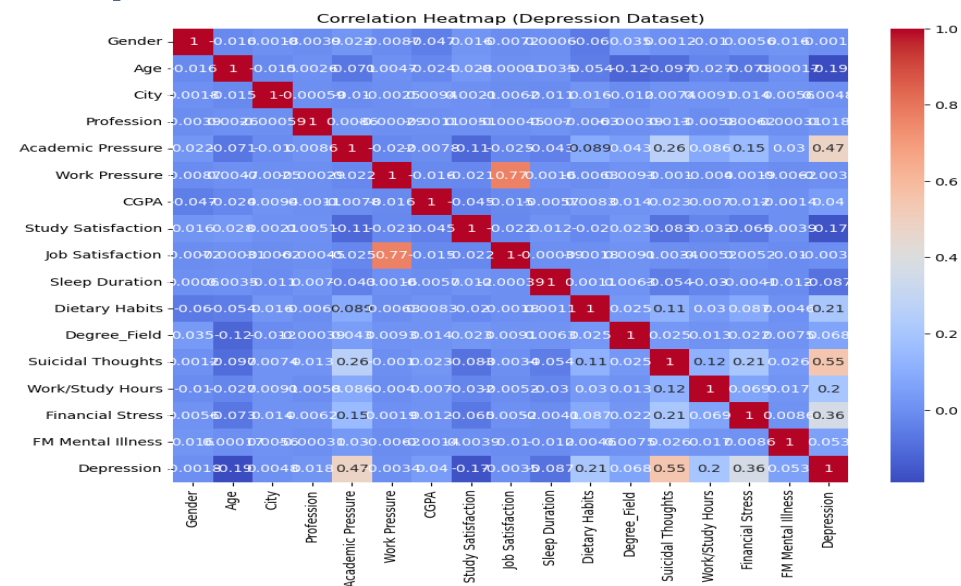
Bullying

0.753

Anxiety

0.753

Depression Predictors - Correlation



Suicidal thoughts

0.546

Academic pressure

0.475

Financial stress

0.364

Short-term psychological & social triggers

Broader psychological, academic and lifestyle influences, reinforcing depression's multifactorial nature.

Limitations & Future Work

Limitations

Datasets & Data Constraints

Self-reported datasets
International datasets (not UK-specific)
Lack of behavioural data

Future Work

Planned Improvements

- UK university datasets
- Deep learning comparison
- Real-time behavioural analytics

Addressing limitations will strengthen real-world applicability and generalisability.

Key Results



ML can predict student mental health risk accurately — Logistic Regression achieves 91.5% on stress, 87.3% on depression. [6, 8]



Simpler models outperform complex ones when data is well-structured — choose models based on data, not assumptions. [6, 7]



Sleep quality, self-esteem, and career anxiety are the most protective/harmful factors — useful for targeted interventions. [1]



Models remain stable under real-world imperfections (missing data, outliers) — ready for practical deployment. [3]



Key Insight: Data quality and interpretability may be more valuable than model complexity in educational mental health prediction. [6]

References

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